# tidy data with tidyr

prof. dr. nicolas meseth

1 what tidy means
2 wide to long
3 long to wide
4 one cell, many things
5 the map

in a tidy workshop every tool has its own place, so you find it without searching, and tidy data gives every value a place of its own in the same way.



image: ai-generated (gpt-image-2.5-flare)

part 1

# what tidy means

one answer, four tables

one row per answer

| respondent | brand | price |
|---|---|---|
| 1 | weihen | 1.49 |
| 1 | gut | 1.09 |
| 1 | alpro | 1.89 |
| 16 | weihen | 1.19 |
| 16 | gut | 1.19 |
| 16 | alpro | 1.49 |
| 36 | weihen | 1.09 |
| 36 | gut | 1.09 |
| 36 | alpro | 1.49 |

one answer, four tables

one row per answer

| respondent | brand | price |
|---|---|---|
| 1 | weihen | 1.49 |
| 1 | gut | 1.09 |
| 1 | alpro | 1.89 |
| 16 | weihen | 1.19 |
| 16 | gut | 1.19 |
| 16 | alpro | 1.49 |
| 36 | weihen | 1.09 |
| 36 | gut | 1.09 |
| 36 | alpro | 1.49 |

one column per brand

| respondent | weihen | gut | alpro |
|---|---|---|---|
| 1 | 1.49 | 1.09 | 1.89 |
| 16 | 1.19 | 1.19 | 1.49 |
| 36 | 1.09 | 1.09 | 1.49 |

one answer, four tables

one row per answer

| respondent | brand | price |
|---|---|---|
| 1 | weihen | 1.49 |
| 1 | gut | 1.09 |
| 1 | alpro | 1.89 |
| 16 | weihen | 1.19 |
| 16 | gut | 1.19 |
| 16 | alpro | 1.49 |
| 36 | weihen | 1.09 |
| 36 | gut | 1.09 |
| 36 | alpro | 1.49 |

one column per brand

| respondent | weihen | gut | alpro |
|---|---|---|---|
| 1 | 1.49 | 1.09 | 1.89 |
| 16 | 1.19 | 1.19 | 1.49 |
| 36 | 1.09 | 1.09 | 1.49 |

all prices in one cell

| respondent | prices |
|---|---|
| 1 | weihen 1.49, gut 1.09, alpro 1.89 |
| 16 | weihen 1.19, gut 1.19, alpro 1.49 |
| 36 | weihen 1.09, gut 1.09, alpro 1.49 |

one answer, four tables

one row per answer

| respondent | brand | price |
|---|---|---|
| 1 | weihen | 1.49 |
| 1 | gut | 1.09 |
| 1 | alpro | 1.89 |
| 16 | weihen | 1.19 |
| 16 | gut | 1.19 |
| 16 | alpro | 1.49 |
| 36 | weihen | 1.09 |
| 36 | gut | 1.09 |
| 36 | alpro | 1.49 |

one column per brand

| respondent | weihen | gut | alpro |
|---|---|---|---|
| 1 | 1.49 | 1.09 | 1.89 |
| 16 | 1.19 | 1.19 | 1.49 |
| 36 | 1.09 | 1.09 | 1.49 |

all prices in one cell

| respondent | prices |
|---|---|
| 1 | weihen 1.49, gut 1.09, alpro 1.89 |
| 16 | weihen 1.19, gut 1.19, alpro 1.49 |
| 36 | weihen 1.09, gut 1.09, alpro 1.49 |

one column per person

| brand | 1 | 16 | 36 |
|---|---|---|---|
| weihen | 1.49 | 1.19 | 1.09 |
| gut | 1.09 | 1.19 | 1.09 |
| alpro | 1.89 | 1.49 | 1.49 |

one answer, four tables

one row per answer

| respondent | brand | price |
|---|---|---|
| 1 | weihen | 1.49 |
| 1 | gut | 1.09 |
| 1 | alpro | 1.89 |
| 16 | weihen | 1.19 |
| 16 | gut | 1.19 |
| 16 | alpro | 1.49 |
| 36 | weihen | 1.09 |
| 36 | gut | 1.09 |
| 36 | alpro | 1.49 |



one column per brand

| respondent | weihen | gut | alpro |
|---|---|---|---|
| 1 | 1.49 | 1.09 | 1.89 |
| 16 | 1.19 | 1.19 | 1.49 |
| 36 | 1.09 | 1.09 | 1.49 |

all prices in one cell

| respondent | prices |
|---|---|
| 1 | weihen 1.49, gut 1.09, alpro 1.89 |
| 16 | weihen 1.19, gut 1.19, alpro 1.49 |
| 36 | weihen 1.09, gut 1.09, alpro 1.49 |

one column per person

| brand | 1 | 16 | 36 |
|---|---|---|---|
| weihen | 1.49 | 1.19 | 1.09 |
| gut | 1.09 | 1.19 | 1.09 |
| alpro | 1.89 | 1.49 | 1.49 |

all four hold the same numbers, but only the first lets you ask a question with the verbs you know.

| respondent | brand | price |
|---|---|---|
| 1 | weihen | 1.49 |
| 1 | gut | 1.09 |
| 1 | alpro | 1.89 |
| 16 | weihen | 1.19 |
| 16 | gut | 1.19 |
| 16 | alpro | 1.49 |
| 36 | weihen | 1.09 |
| 36 | gut | 1.09 |
| 36 | alpro | 1.49 |

| respondent | brand | price |
|---|---|---|
| 1 | weihen | 1.49 |
| 1 | gut | 1.09 |
| 1 | alpro | 1.89 |
| 16 | weihen | 1.19 |
| 16 | gut | 1.19 |
| 16 | alpro | 1.49 |
| 36 | weihen | 1.09 |
| 36 | gut | 1.09 |
| 36 | alpro | 1.49 |

every variable has its own column

| respondent | brand | price |
|---|---|---|
| 1 | weihen | 1.49 |
| 1 | gut | 1.09 |
| 1 | alpro | 1.89 |
| 16 | weihen | 1.19 |
| 16 | gut | 1.19 |
| 16 | alpro | 1.49 |
| 36 | weihen | 1.09 |
| 36 | gut | 1.09 |
| 36 | alpro | 1.49 |

every observation has its own row

| respondent | brand | price |
|---|---|---|
| 1 | weihen | 1.49 |
| 1 | gut | 1.09 |
| 1 | alpro | 1.89 |
| 16 | weihen | 1.19 |
| 16 | gut | 1.19 |
| 16 | alpro | 1.49 |
| 36 | weihen | 1.09 |
| 36 | gut | 1.09 |
| 36 | alpro | 1.49 |

every value has its own cell

what is one observation?

| respondent | age | weihen | gut | alpro |
|---|---|---|---|---|
| 1 | 55 | 1.49 | 1.09 | 1.89 |
| 16 | 46 | 1.19 | 1.19 | 1.49 |
| 36 | 53 | 1.09 | 1.09 | 1.49 |

| respondent | brand | price |
|---|---|---|
| 1 | weihen | 1.49 |
| 1 | gut | 1.09 |
| 1 | alpro | 1.89 |
| 16 | weihen | 1.19 |
| 16 | gut | 1.19 |
| 16 | alpro | 1.49 |
| 36 | weihen | 1.09 |
| 36 | gut | 1.09 |
| 36 | alpro | 1.49 |

what is one observation?

how old are the people who answered?

| respondent | age | weihen | gut | alpro |
|---|---|---|---|---|
| 1 | 55 | 1.49 | 1.09 | 1.89 |
| 16 | 46 | 1.19 | 1.19 | 1.49 |
| 36 | 53 | 1.09 | 1.09 | 1.49 |



| respondent | brand | price |
|---|---|---|
| 1 | weihen | 1.49 |
| 1 | gut | 1.09 |
| 1 | alpro | 1.89 |
| 16 | weihen | 1.19 |
| 16 | gut | 1.19 |
| 16 | alpro | 1.49 |
| 36 | weihen | 1.09 |
| 36 | gut | 1.09 |
| 36 | alpro | 1.49 |

what is one observation?

how old are the people who answered?

| respondent | age | weihen | gut | alpro |
|---|---|---|---|---|
| 1 | 55 | 1.49 | 1.09 | 1.89 |
| 16 | 46 | 1.19 | 1.19 | 1.49 |
| 36 | 53 | 1.09 | 1.09 | 1.49 |



which brand is thought to be cheapest?

| respondent | brand | price |
|---|---|---|
| 1 | weihen | 1.49 |
| 1 | gut | 1.09 |
| 1 | alpro | 1.89 |
| 16 | weihen | 1.19 |
| 16 | gut | 1.19 |
| 16 | alpro | 1.49 |
| 36 | weihen | 1.09 |
| 36 | gut | 1.09 |
| 36 | alpro | 1.49 |



what counts as one observation depends on the question, so tidy is a property of a table and a question together.

## why it pays off

one column per brand

```
prices |>
  summarize(
    weihen = median(weihen),
    gut = median(gut),
    alpro = median(alpro)
  )
```

one line for every brand

one row per person and brand

```
prices_long |>
  group_by(brand) |>
  summarize(median = median(price))
```

one line for all brands

in a tidy table one line of code covers every brand, however many there are.

respondents gave a price for each of three brands, one column per brand. is this table tidy?

yes, one row per person

no, values in column names

no, two values in one cell

it depends on the question

live.hsos.uber.space/q

respondents gave a price for each of three brands, one column per brand. is this table tidy?

- yes, one row per person
- no, values in column names
- no, two values in one cell
- it depends on the question

live.hsos.uber.space/q

respondents gave a price for each of three brands, one column per brand. is this table tidy?

- yes, one row per person
- no, values in column names
- no, two values in one cell
- it depends on the question



live.hsos.uber.space/q

values in column names

| respondent | weihen | gut | alpro |
|---|---|---|---|
| 1 | 1.49 | 1.09 | 1.89 |
| 16 | 1.19 | 1.19 | 1.49 |

`pivot_longer()`

two variables in one column

| respondent | measure | value |
|---|---|---|
| 1 | price | 1.49 |
| 1 | sure | 2 |
| 16 | price | 1.19 |

`pivot_wider()`

two values in one cell

| brand | range |
|---|---|
| weihen | 1.30 to 1.50 |
| gut | 0.90 to 1.10 |

`separate_wider_delim()`

a list in one cell

| respondent | shops |
|---|---|
| 12 | supermarket |
| 24 | discounter, supermarket, farm shop |

`separate_longer_delim()`

part 2

# wide to long

```
library(tidyverse)

raw <- read_csv(
  "https://hands-on-computer-science.de/media/datasets/mds12_schoko_milch.csv"
) |>
  mutate(respondent = row_number(), .before = 1)
```

one row per respondent, 2,811 rows and 606 columns, but no column that tells the respondents apart, so `row_number()` gives every row a number in the new column `respondent`

the price question, as the survey stores it

| respondent | u013pzahl_1weihen | u013pzahl_2gut | u013pzahl_3alpro |
|---|---|---|---|
| 1 | 1.49 | 1.09 | 1.89 |
| 2 | NA | NA | NA |
| 3 | NA | NA | NA |
| … | … | … | … |
| 16 | 1.19 | 1.19 | 1.49 |
| … | … | … | … |
| 36 | 1.09 | 1.09 | 1.49 |

2,580 of the 2,811 rows are empty in all three columns,
because only 231 people were asked this question

the brand is a variable, but here it hides in three column names.

pivot_longer(): columns become rows

| respondent | weihen | gut | alpro |
|---|---|---|---|
| 1 | 1.49 | 1.09 | 1.89 |
| 16 | 1.19 | 1.19 | 1.49 |
| 36 | 1.09 | 1.09 | 1.49 |

pivot_longer(): columns become rows

| respondent | weihen | gut | alpro |
|---|---|---|---|
| 1 | 1.49 | 1.09 | 1.89 |
| 16 | 1.19 | 1.19 | 1.49 |
| 36 | 1.09 | 1.09 | 1.49 |

pivot_longer(): columns become rows

| respondent | weihen | gut | alpro |
|---|---|---|---|
| 1 | 1.49 | 1.09 | 1.89 |
| 16 | 1.19 | 1.19 | 1.49 |
| 36 | 1.09 | 1.09 | 1.49 |

| respondent | brand | price |
|---|---|---|
| 1 | weihen | 1.49 |
| 1 | gut | 1.09 |
| 1 | alpro | 1.89 |

pivot_longer(): columns become rows

| respondent | weihen | gut | alpro |
|---|---|---|---|
| 1 | 1.49 | 1.09 | 1.89 |
| 16 | 1.19 | 1.19 | 1.49 |
| 36 | 1.09 | 1.09 | 1.49 |

| respondent | brand | price |
|---|---|---|
| 1 | weihen | 1.49 |
| 1 | gut | 1.09 |
| 1 | alpro | 1.89 |
| 16 | weihen | 1.19 |
| 16 | gut | 1.19 |
| 16 | alpro | 1.49 |
| 36 | weihen | 1.09 |
| 36 | gut | 1.09 |
| 36 | alpro | 1.49 |

pivot_longer(): columns become rows

| | names_to | values_to |
|---|---|---|
| respondent | brand | price |
| 1 | weihen | 1.49 |
| 1 | gut | 1.09 |
| 1 | alpro | 1.89 |
| 16 | weihen | 1.19 |
| 16 | gut | 1.19 |
| 16 | alpro | 1.49 |
| 36 | weihen | 1.09 |
| 36 | gut | 1.09 |
| 36 | alpro | 1.49 |

```
prices <- tribble(
  ~respondent, ~weihen, ~gut, ~alpro,
  1,           1.49,    1.09, 1.89,
  16,          1.19,    1.19, 1.49,
  36,          1.09,    1.09, 1.49
)

prices |>
  pivot_longer(
    cols = c(weihen, gut, alpro),
    names_to = "brand",
    values_to = "price"
  )
```

which columns?
where do their names go?
where do their values go?

with the real survey

```
raw |>
  select(respondent, age = Q002alter, starts_with("u013pzahl")) |>
  pivot_longer(
    cols = starts_with("u013pzahl"),
    names_to = "brand",
    values_to = "price"
  )
```

with the real survey

```
raw |>
  select(respondent, age = Q002alter, starts_with("u013pzahl")) |>
  pivot_longer(
    cols = starts_with("u013pzahl"),
    names_to = "brand",
    values_to = "price"
  )
```

```
# A tibble: 8,433 × 4
```

| respondent | age | brand | price |
|---|---|---|---|
| 1 | 55 | u013pzahl_1weihen | 1.49 |
| 1 | 55 | u013pzahl_2gut | 1.09 |
| 1 | 55 | u013pzahl_3alpro | 1.89 |
| 2 | 35 | u013pzahl_1weihen | NA |
| 2 | 35 | u013pzahl_2gut | NA |
| 2 | 35 | u013pzahl_3alpro | NA |

## cleaning up on the way

```
prices_long <- raw |>
  select(respondent, age = Q002alter, starts_with("u013pzahl")) |>
  pivot_longer(
    cols = starts_with("u013pzahl"),
    names_to = "brand",
    names_pattern = "u013pzahl_\\d(.*)",
    values_to = "price",
    values_drop_na = TRUE
  )
```

| | |
|---|---|
| `names_prefix` | removes a fixed start of the name |
| `names_pattern` | keeps what the brackets catch |
| `names_sep` | splits the name at a character |
| `values_drop_na` | drops the empty values |

more arguments of pivot_longer()

## cleaning up on the way

```
prices_long <- raw |>
  select(respondent, age = Q002alter, starts_with("u013pzahl")) |>
  pivot_longer(
    cols = starts_with("u013pzahl"),
    names_to = "brand",
    names_pattern = "u013pzahl_\\d(.*)",
    values_to = "price",
    values_drop_na = TRUE
  )
```

| | |
|---|---|
| `names_prefix` | removes a fixed start of the name |
| `names_pattern` | keeps what the brackets catch |
| `names_sep` | splits the name at a character |
| `values_drop_na` | drops the empty values |

more arguments of pivot_longer()

```
# A tibble: 693 × 4
```

| respondent | age | brand | price |
|---|---|---|---|
| 1 | 55 | weihen | 1.49 |
| 1 | 55 | gut | 1.09 |
| 1 | 55 | alpro | 1.89 |

now one line answers it

```
prices_long |>
  filter(price <= 10) |>
  group_by(brand) |>
  summarize(n = n(), median = median(price))
```

now one line answers it

```
prices_long |>
  filter(price <= 10) |>
  group_by(brand) |>
  summarize(n = n(), median = median(price))
```

| brand | n | median |
|---|---|---|
| alpro | 224 | 1.6 |
| gut | 226 | 1.09 |
| weihen | 225 | 1.49 |

respondents find a higher price fair for alpro than for weihenstephan, although the survey showed both at 1.79.

nine columns of zeros and ones

| respondent | v008ort_1discount | v008ort_2super | v008ort_3sbmarkt | v008ort_4fach |
|---|---|---|---|---|
| 1 | 1 | 1 | 0 | 0 |
| 2 | 1 | 1 | 0 | 1 |
| 3 | 1 | 1 | 0 | 0 |
| 4 | 1 | 0 | 1 | 0 |

```
# 5 more variables: v008ort_5markt, v008ort_6hof,
#   v008ort_7bioladen, v008ort_8onlineds, v008ort_9online
```

a question with several answers is one variable, however many columns the survey tool makes of it.

```
raw |>
  select(respondent, starts_with("v008ort")) |>
  pivot_longer(
    cols = starts_with("v008ort"),
    names_to = "shop",
    names_prefix = "v008ort_",
    values_to = "value"
  ) |>
  filter(value == 1) |>
  count(shop)
```

```
raw |>
  select(respondent, starts_with("v008ort")) |>
  pivot_longer(
    cols = starts_with("v008ort"),
    names_to = "shop",
    names_prefix = "v008ort_",
    values_to = "value"
  ) |>
  filter(value == 1) |>
  count(shop)
```

| shop | n |
|---|---|
| 1discount | 2065 |
| 2super | 2134 |
| 3sbmarkt | 848 |
| 4fach | 1029 |
| 5markt | 493 |
| 6hof | 396 |
| 7bioladen | 359 |
| 8onlineds | 303 |
| 9online | 173 |

part 3

# long to wide

two variables in one column

| respondent | brand | measure | value |
|---|---|---|---|
| 1 | weihen | price | 1.49 |
| 1 | weihen | sure | 2 |
| 1 | alpro | price | 1.89 |
| 1 | alpro | sure | -1 |
| 16 | weihen | price | 1.19 |
| 16 | weihen | sure | 3 |
| 16 | alpro | price | 1.49 |
| 16 | alpro | sure | 1 |

price and certainty are two variables, so they need two columns, even if the table looks tidy at first sight.

pivot_wider(): rows become columns

| respondent | brand | measure | value |
|---|---|---|---|
| 1 | weihen | price | 1.49 |
| 1 | weihen | sure | 2 |
| 1 | alpro | price | 1.89 |
| 1 | alpro | sure | -1 |
| 16 | weihen | price | 1.19 |
| 16 | weihen | sure | 3 |
| 16 | alpro | price | 1.49 |
| 16 | alpro | sure | 1 |

pivot_wider(): rows become columns

| respondent | brand | measure | value |
|---|---|---|---|
| 1 | weihen | price | 1.49 |
| 1 | weihen | sure | 2 |
| 1 | alpro | price | 1.89 |
| 1 | alpro | sure | -1 |
| 16 | weihen | price | 1.19 |
| 16 | weihen | sure | 3 |
| 16 | alpro | price | 1.49 |
| 16 | alpro | sure | 1 |

pivot_wider(): rows become columns

| respondent | brand | measure | value |
|---|---|---|---|
| 1 | weihen | price | 1.49 |
| 1 | weihen | sure | 2 |
| 1 | alpro | price | 1.89 |
| 1 | alpro | sure | -1 |
| 16 | weihen | price | 1.19 |
| 16 | weihen | sure | 3 |
| 16 | alpro | price | 1.49 |
| 16 | alpro | sure | 1 |

| respondent | brand | price | sure |
|---|---|---|---|
| 1 | weihen | 1.49 | 2 |

pivot_wider(): rows become columns

| respondent | brand | measure | value |
|---|---|---|---|
| 1 | weihen | price | 1.49 |
| 1 | weihen | sure | 2 |
| 1 | alpro | price | 1.89 |
| 1 | alpro | sure | -1 |
| 16 | weihen | price | 1.19 |
| 16 | weihen | sure | 3 |
| 16 | alpro | price | 1.49 |
| 16 | alpro | sure | 1 |

| respondent | brand | price | sure |
|---|---|---|---|
| 1 | weihen | 1.49 | 2 |
| 1 | alpro | 1.89 | -1 |
| 16 | weihen | 1.19 | 3 |
| 16 | alpro | 1.49 | 1 |

the code

```
answers <- tribble(
  ~respondent, ~brand,   ~measure, ~value,
  1,           "weihen", "price",  1.49,
  1,           "weihen", "sure",   2,
  1,           "alpro",  "price",  1.89,
  1,           "alpro",  "sure",   -1,
  16,          "weihen", "price",  1.19,
  16,          "weihen", "sure",   3,
  16,          "alpro",  "price",  1.49,
  16,          "alpro",  "sure",   1
)

answers |>
  pivot_wider(
    names_from = measure,
    values_from = value
  )
```

which column gives the new names?
which column fills them?

a long table has 3 people, 3 brands and 2 measures in rows. how many rows does it have?

6

9

18

27



live.hsos.uber.space/q

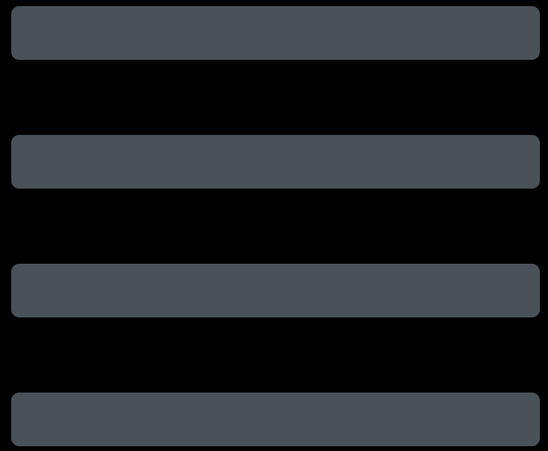
a long table has 3 people, 3 brands and 2 measures in rows. how many rows does it have?

6

9

18

27



live.hsos.uber.space/q

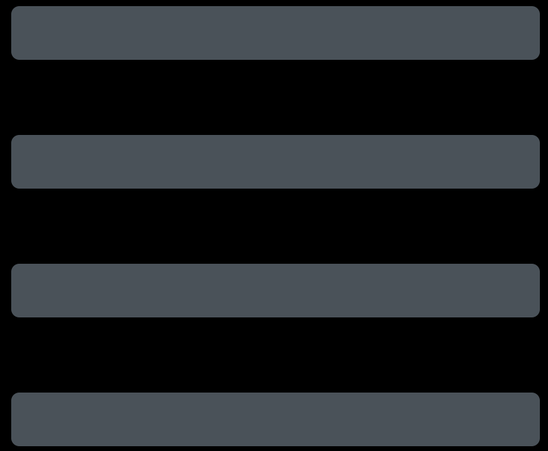
a long table has 3 people, 3 brands and 2 measures in rows. how many rows does it have?

6

9

18

27



live.hsos.uber.space/q

```
prices_long |>
  filter(price <= 10) |>
  mutate(decade = age %/% 10 * 10) |>
  group_by(decade, brand) |>
  summarize(median = median(price)) |>
  pivot_wider(
    names_from = brand,
    values_from = median
  )
```

keep data long while you analyse it, and make it wide only when a person has to read it.

```
prices_long |>
  filter(price <= 10) |>
  mutate(decade = age %/% 10 * 10) |>
  group_by(decade, brand) |>
  summarize(median = median(price)) |>
  pivot_wider(
    names_from = brand,
    values_from = median
  )
```

| decade | alpro | gut | weihen |
|---|---|---|---|
| 10 | 1.80 | 1.50 | 1.80 |
| 20 | 1.80 | 1.10 | 1.80 |
| 30 | 1.69 | 1.00 | 1.40 |
| 40 | 1.50 | 1.09 | 1.35 |
| 50 | 1.60 | 1.09 | 1.50 |
| 60 | 1.59 | 1.09 | 1.39 |
| 70 | 1.40 | 1.11 | 1.20 |

keep data long while you analyse it, and make it wide only when a person has to read it.

| respondent | weihen | gut | alpro |
|---|---|---|---|
| 1 | 1.49 | 1.09 | 1.89 |
| 16 | 1.19 | 1.19 | 1.49 |
| 36 | 1.09 | 1.09 | 1.49 |

pivot_longer()

pivot_wider()

| respondent | brand | price |
|---|---|---|
| 1 | weihen | 1.49 |
| 1 | gut | 1.09 |
| 1 | alpro | 1.89 |
| 16 | weihen | 1.19 |
| 16 | gut | 1.19 |
| 16 | alpro | 1.49 |
| 36 | weihen | 1.09 |
| 36 | gut | 1.09 |
| 36 | alpro | 1.49 |

two variables across two brands

| respondent | price_weihen | price_alpro | sure_weihen | sure_alpro |
|---|---|---|---|---|
| 1 | 1.49 | 1.89 | 2 | -1 |
| 16 | 1.19 | 1.49 | 3 | 1 |

two variables across two brands

| respondent | price weihen | price alpro | sure weihen | sure alpro |
|---|---|---|---|---|
| 1 | 1.49 | 1.89 | 2 | -1 |
| 16 | 1.19 | 1.49 | 3 | 1 |

two variables across two brands

| respondent | brand | price | sure |
|---|---|---|---|
| 1 | weihen | 1.49 | 2 |
| 1 | alpro | 1.89 | -1 |
| 16 | weihen | 1.19 | 3 |
| 16 | alpro | 1.49 | 1 |

names_to with ".value"

```
both <- tribble(
  ~respondent, ~price_weihen, ~price_alpro, ~sure_weihen, ~sure_alpro,
  1,           1.49,          1.89,         2,            -1,
  16,          1.19,          1.49,         3,            1
)

both |>
  pivot_longer(
    cols = -respondent,
    names_to = c(".value", "brand"),
    names_sep = "_"
  )
```

the result is the tidy table with the columns `respondent`, `brand`, `price` and `sure`

part 4

# one cell, many things

| brand | range |
|---|---|
| weihen | 1.30 to 1.50 |
| gut | 0.90 to 1.10 |
| alpro | 1.50 to 1.70 |

| brand | range |
|---|---|
| weihen | 1.30 to 1.50 |
| gut | 0.90 to 1.10 |
| alpro | 1.50 to 1.70 |

| brand | from | to |
|---|---|---|
| weihen | 1.30 | 1.50 |
| gut | 0.90 | 1.10 |
| alpro | 1.50 | 1.70 |

```
ranges <- tribble(
  ~brand,   ~range,
  "weihen", "1.30 to 1.50",
  "gut",    "0.90 to 1.10",
  "alpro",  "1.50 to 1.70"
)

ranges |>
  separate_wider_delim(
    range,
    delim = " to ",
    names = c("from", "to")
  ) |>
  mutate(from = as.numeric(from), to = as.numeric(to))
```

a list in one cell

| respondent | shops |
|---|---|
| 12 | supermarket |
| 24 | discounter, supermarket, farm shop |
| 48 | supermarket, specialist shop, organic shop |

a list in one cell

| respondent | shops |
|---|---|
| 12 | supermarket |
| 24 | discounter, supermarket, farm shop |
| 48 | supermarket, specialist shop, organic shop |

# a list in one cell

| respondent | shops |
|---|---|
| 12 | supermarket |
| 24 | discounter |
| 24 | supermarket |
| 24 | farm shop |
| 48 | supermarket |
| 48 | specialist shop |
| 48 | organic shop |

```
shopping <- tribble(
  ~respondent, ~shops,
  12,          "supermarket",
  24,          "discounter, supermarket, farm shop",
  48,          "supermarket, specialist shop, organic shop"
)

shopping |>
  separate_longer_delim(shops, delim = ", ") |>
  count(shops)
```

wider splits a cell into columns, longer splits it into rows, and the question decides which one you need.

```
shopping <- tribble(
  ~respondent, ~shops,
  12,          "supermarket",
  24,          "discounter, supermarket, farm shop",
  48,          "supermarket, specialist shop, organic shop"
)

shopping |>
  separate_longer_delim(shops, delim = ", ") |>
  count(shops)
```

| shops | n |
|---|---|
| discounter | 1 |
| farm shop | 1 |
| organic shop | 1 |
| specialist shop | 1 |
| supermarket | 3 |

wider splits a cell into columns, longer splits it into rows, and the question decides which one you need.

# one question, three shapes

a list in one cell

| respondent | shops |
|---|---|
| 12 | supermarket |
| 24 | discounter, supermarket, farm shop |
| 48 | supermarket, specialist shop, organic shop |

`separate_longer_delim()`

tidy: one row per person and shop

| respondent | shop |
|---|---|
| 12 | supermarket |
| 24 | discounter |
| 24 | supermarket |
| 24 | farm shop |
| 48 | supermarket |
| 48 | specialist shop |
| 48 | organic shop |

# one question, three shapes

a list in one cell

| respondent | shops |
|---|---|
| 12 | supermarket |
| 24 | discounter, supermarket, farm shop |
| 48 | supermarket, specialist shop, organic shop |

separate_longer_delim()

one column per shop, 1 means ticked

| respondent | discounter | supermarket | specialist shop | farm shop | organic shop | + 4 more |
|---|---|---|---|---|---|---|
| 12 | 0 | 1 | 0 | 0 | 0 | |
| 24 | 1 | 1 | 0 | 1 | 0 | |
| 48 | 0 | 1 | 1 | 0 | 1 | |

pivot_longer() |>
filter()

tidy: one row per person and shop

| respondent | shop |
|---|---|
| 12 | supermarket |
| 24 | discounter |
| 24 | supermarket |
| 24 | farm shop |
| 48 | supermarket |
| 48 | specialist shop |
| 48 | organic shop |

a cell holds "vegan, organic, regional" and you want to count how often each label occurs. which verb?

pivot_longer()

pivot_wider()

separate_wider_delim()

separate_longer_delim()



live.hsos.uber.space/q

a cell holds "vegan, organic, regional" and you want to count how often each label occurs. which verb?

- pivot_longer()
- pivot_wider()
- separate_wider_delim()
- separate_longer_delim()

live.hsos.uber.space/q

a cell holds "vegan, organic, regional" and you want to count how often each label occurs. which verb?

- pivot_longer()
- pivot_wider()
- separate_wider_delim()
- separate_longer_delim()

live.hsos.uber.space/q

part 5

# the map

in a spreadsheet a group is often named only once, at the top of its block, because the eye fills in the rest.

| brand | respondent | price |
|---|---|---|
| weihen | 1 | 1.49 |
| | 16 | 1.19 |
| | 36 | 1.09 |
| alpro | 1 | 1.89 |
| | 16 | 1.49 |
| | 36 | 1.49 |

```
sheet <- tribble(
  ~brand,   ~respondent, ~price,
  "weihen", 1,           1.49,
  NA,       16,          1.19,
  NA,       36,          1.09,
  "alpro",  1,           1.89,
  NA,       16,          1.49,
  NA,       36,          1.49
)

sheet |>
  fill(brand)
```

| brand | respondent | price |
|---|---|---|
| weihen | 1 | 1.49 |
| NA | 16 | 1.19 |
| NA | 36 | 1.09 |
| alpro | 1 | 1.89 |
| NA | 16 | 1.49 |
| NA | 36 | 1.49 |

```
sheet <- tribble(
  ~brand,   ~respondent, ~price,
  "weihen", 1,           1.49,
  NA,       16,          1.19,
  NA,       36,          1.09,
  "alpro",  1,           1.89,
  NA,       16,          1.49,
  NA,       36,          1.49
)

sheet |>
  fill(brand)
```

| brand | respondent | price |
|---|---|---|
| weihen | 1 | 1.49 |
| weihen | 16 | 1.19 |
| weihen | 36 | 1.09 |
| alpro | 1 | 1.89 |
| alpro | 16 | 1.49 |
| alpro | 36 | 1.49 |

from symptom to verb

| what you see | what you do |
|---|---|
| values in column names | `pivot_longer()` |
| two variables in one column | `pivot_wider()` |
| two values in one cell | `separate_wider_delim()` |
| a list in one cell | `separate_longer_delim()` |
| a group named only once | `fill()` |

tidying comes first and
costs time, but after it
every new question is only
a few lines of code.

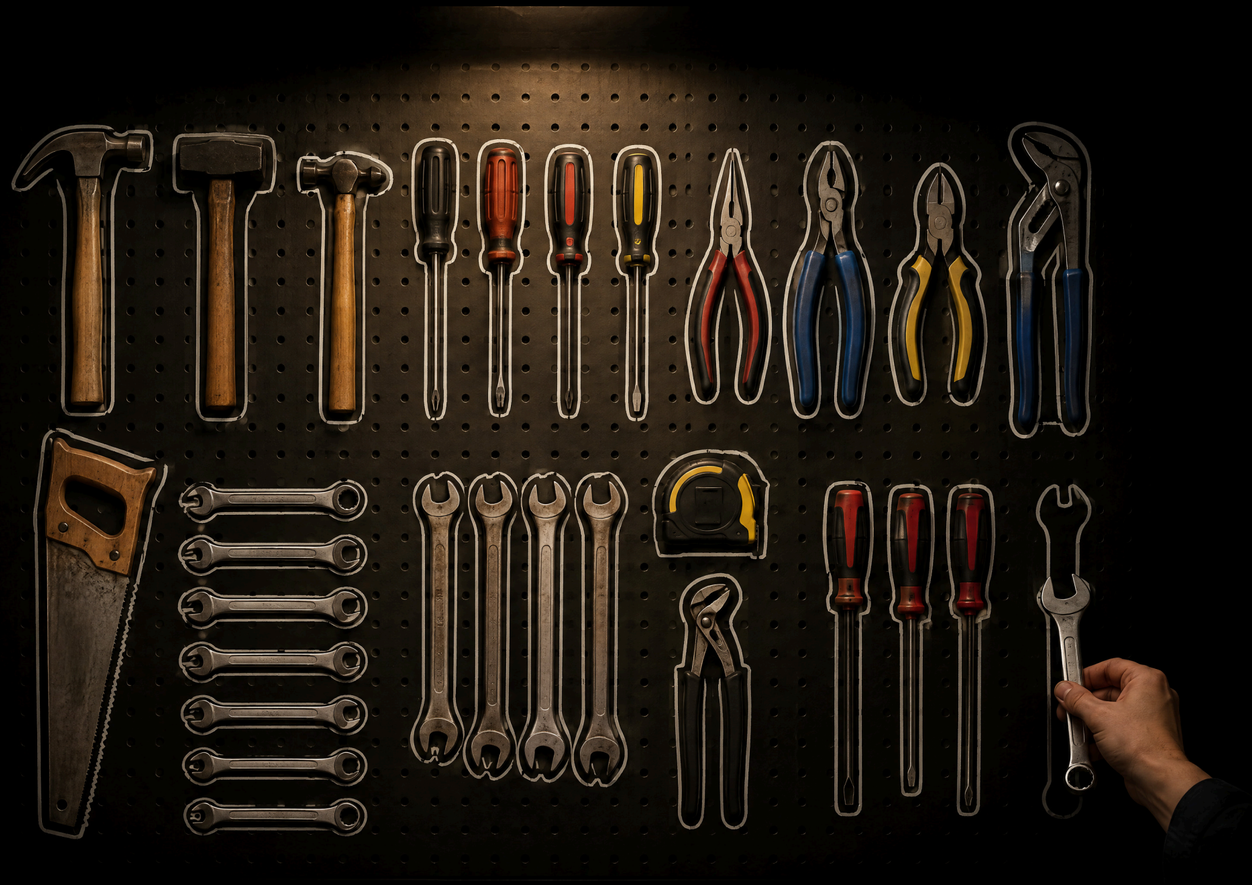


image: ai-generated (gpt-image-2.5-flare)